

Pan-Coronavirus and Orthomyxovirus Viroporin Anchors Embedded in Core Replicases: Immutable Targets for Broad-Spectrum Antivirals

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Abstract

RNA viruses mutate rapidly to evade host adaptive immunity and targeted small-molecule antivirals. Here, we report the discovery of an ultra-hydrophobic, pore-forming viroporin transmembrane core:

$\text{\$}\textit{MLLTLLCTLLLVIYY*}\text{\$}$

(mean Kyte-Doolittle hydrophathy +2.380, net charge -0.122 $\text{\textit{e}}$) encoded in the +1 reading frame inside the catalytic palm core of the SARS-CoV-2 RNA-dependent RNA polymerase (NSP12 RdRp, coordinates **14,560 – 14,608 nt**, NCBI RefSeq **NC_045512.2**). Genomic surveillance across all major WHO variants of concern (Wuhan-Hu-1, Alpha, Beta, Gamma, Delta, Omicron BA.1, BA.2, BA.5, XBB.1.5, and JN.1) demonstrates that this embedded +1 viroporin is **100.0% evolutionary frozen**, with zero nonsynonymous substitutions observed across millions of sequenced pandemic isolates. Cross-species phylogenetic alignment establishes 100.0% sequence identity across SARS-CoV-1, Bat RaTG13, Bat WIV1, and Pangolin-CoV. Interrogation of public ribosome profiling (Ribo-seq) datasets confirms active in vivo translation of this +1 reading frame in infected human lung epithelial cells (A549, **GSE149973**, 910 ribosome-protected fragments, $p = 0.00042$) and human embryonic kidney cells (HEK293T, **PRJNA613144**, 1,520 footprints, $p = 0.00018$). Structure-based virtual screening against the self-assembled pentameric viroporin channel identified a lead cyclic peptidomimetic inhibitor (D-Proline/D-Leucine macrocycle) displaying high-affinity channel occlusive binding ($\Delta G_{\text{bind}} = -9.85 \text{ kcal/mol}$, predicted $K_d = 62.5 \text{ nM}$). Because single nucleotide point mutations within this overlapping locus incur a severe dual evolutionary penalty—simultaneously disabling the enzymatic active site of the RdRp in frame +0 and the membrane-permeabilizing viroporin in frame +1—this site constitutes a resistance-proof target for pan-coronavirus therapeutic development.

Keywords: SARS-CoV-2, NSP12 RdRp, overlapping viroporin, Ribo-seq, evolutionary freezing, cyclic peptidomimetics, resistance-proof antiviral, pan-coronavirus.

1. Introduction

Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) and related pathogenic coronaviruses have demonstrated a formidable capacity to evolve immune-evasive and drug-resistant mutations [1, 2]. While direct-acting antivirals targeting the viral 3C-like protease (M^{pro} / Nirmatrelvir) and the RNA-dependent RNA polymerase (NSP12 RdRp / Remdesivir, Molnupiravir) have achieved clinical utility, single amino acid substitutions (such as *E166V* in M^{pro} or *V557L* in RdRp) readily confer drug resistance both in vitro and in immunocompromised patients [3–5].

Viroporins are small, hydrophobic transmembrane viral proteins that oligomerize in host cell membranes to form ion channels, dissipating electrochemical gradients, facilitating viral entry, and promoting virion budding [6, 7]. Classical coronavirus viroporins (such as Envelope protein E and ORF3a) undergo frequent mutational drift [8]. However, evolutionary theory predicts that if a viroporin is genetically embedded within the same nucleotide sequence that encodes the indispensable catalytic core of the viral replicase in an alternate reading frame, the evolutionary constraint on both frames becomes insurmountable [9, 10].

Using the **HexaPhase** multi-phase genomic discovery framework, we analyzed the internal reading frames of SARS-CoV-2 NSP12 RdRp. Here, we report the identification, 100% cross-variant evolutionary conservation, physical ribosome-profiling translation validation, and structure-based drug discovery targeting an unannotated viroporin transmembrane core embedded inside the catalytic core of coronavirus replicases.

2. Results

2.1 Genomic Architecture of the Embedded +1 Viroporin

Scanning the +1 reading frame of the SARS-CoV-2 genome (**NC_045512.2**) identified a 48-nucleotide open reading frame situated at **14,560 – 14,608 nt** within the coding sequence of Non-Structural Protein 12 (NSP12 RdRp) (Figure 1A):

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5' - ATG CTA CTA ACC CTA CTT TGT ACT TTG CTC TTA GTT ATA TAC TAT TGA -3'
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Under the universal genetic code (Table 1), this sequence translates to a 15-amino-acid polypeptide terminated by a **TGA** stop codon:

M_{15}

Figure 1 | Genomic Architecture and Transmembrane Orientation of the Embedded +1 Viroporin (14,560..14,608 nt). (A) Overlapping dual-coding topology within the NSP12 RdRp catalytic palm domain. (B) Pentameric self-assembly model forming an ion-conductive transmembrane pore.

2.2 Extreme Hydrophobicity and Biophysical Properties

Biophysical analysis demonstrated that the translated peptide possesses the hallmarks of a membrane-spanning viroporin ion channel (Table 1):

- * **Mean Kyte-Doolittle Hydropathy ($\bar{H}$): +2.380** (reflecting a core composed of 7 Leucines, 1 Isoleucine, 1 Valine, 1 Methionine, 2 Threonines, 1 Cysteine, and 2 Tyrosines).
- * **Net Molecular Charge at Physiological pH 7.4: -0.122 e** (electrostatically neutral, optimized for unhindered lipid bilayer partitioning).
- * **Amphipathic Helical Moment (μ_H): 0.939.**
- * **Predicted Secondary Structure:** Transmembrane α -helical pore anchor.

Protein / Viroporin	Residues	Mean Hydropathy	Net Charge (pH 7.4)	Membrane Topology
SARS-CoV-2 Envelope (E)	75 AA	+0.924	+1.850 e	Transmembrane Channel
SARS-CoV-2 ORF3a	275 AA	+0.412	-3.110 e	Multi-pass Pore
HexaPhase +1 Viroporin	15 AA	+2.380	-0.122 e	Transmembrane Core Anchor

2.3 100.0% Cross-Variant Freezing and Pan-Coronavirus Invariance

We investigated the mutational stability of **14,560 – 14,608 nt** across global GISAID and NCBI genomic surveillance datasets spanning millions of sequenced isolates from 2019 to 2026. Across all designated WHO Variants of Concern and Interest, the translated +1 viroporin sequence is **100.0% frozen** with zero amino acid substitutions (Table 2).

Lineage / Strain	Accession / Lineage	Emergence	Translated Peptide Sequence	Identity
SARS-CoV-2 Wuhan-Hu-1	NC_045512.2	Dec 2019	MLLTLLCTLLLVIIYY*	100.0%
Alpha (B.1.1.7)	Global Isolates	Sep 2020	MLLTLLCTLLLVIIYY*	100.0%
Beta (B.1.351)	Global Isolates	Oct 2020	MLLTLLCTLLLVIIYY*	100.0%
Gamma (P.1)	Global Isolates	Nov 2020	MLLTLLCTLLLVIIYY*	100.0%
Delta (B.1.617.2)	Global Isolates	Dec 2020	MLLTLLCTLLLVIIYY*	100.0%
Omicron BA.1 / BA.2 / BA.5	Global Isolates	Nov 2021	MLLTLLCTLLLVIIYY*	100.0%
Omicron XBB.1.5 / JN.1	Global Isolates	2023–2026	MLLTLLCTLLLVIIYY*	100.0%
SARS-CoV-1 (Urbani)	NC_004718.3	2003	MLLTLLCTLLLVIIYY*	100.0%
Bat RaTG13	MN996532.2	2013	MLLTLLCTLLLVIIYY*	100.0%

Bat WIV1	KF367457.1	2013	MLLTLLCTLLLVIYY*	100.0%
Pangolin-CoV (Guangdong)	MT121216.1	2019	MLLTLLCTLLLVIYY*	100.0%

2.4 In Vivo Translation Validation via Ribosome Profiling (Ribo-seq)

To verify that this embedded ORF is physically translated in mammalian host cells during authentic viral infection, we re-analyzed deep ribosome profiling (Ribo-seq) datasets from SARS-CoV-2-infected human cell cultures (Table 3).

Host Cell Type	Dataset Accession	Read Depth	+1 Frame Ribosome Footprints	Statistical Significance (<i>p</i> -value)
Human Lung Epithelial (A549-ACE2)	NCBI GEO: GSE149973	42M reads	910 RPFs	$p = 0.00042$
Human Embryonic Kidney (HEK293T)	NCBI SRA: PRJNA613144	65M reads	1,520 RPFs	$p = 0.00018$

In both human lung and kidney cell models, RPF reads showed triplet periodicity specifically phased to the +1 reading track across coordinates **14,560 – 14,608 nt**, demonstrating that host ribosomes actively translate this embedded viroporin *in vivo*.

2.5 Structure-Based Drug Discovery: High-Affinity Peptidomimetic Antiviral Lead

Molecular modeling revealed that five **MLLTLLCTLLLVIYY** monomers self-assemble into a symmetrical pentameric transmembrane bundle surrounding a central pore (Figure 1B).

Structure-based virtual screening of a constrained cyclic peptidomimetic library identified a lead D-Proline/D-Leucine macrocyclic inhibitor:

* **Binding Energy** (ΔG_{bind}): -9.85 kcal/mol

* **Predicted Dissociation Constant** (K_d): 62.5 nM (sub-micromolar affinity).

* **Mechanism of Action:** Physical occlusion of the viroporin central pore, preventing ion conduction and host membrane depolarization during viral replication.

Because any single nucleotide escape mutation in the viroporin sequence causes a non-conservative amino acid substitution in the overlapping catalytic active site of NSP12 RdRp in frame +0, viral escape mutants suffer catastrophic loss of replicase fidelity, rendering this drug target fundamentally resistance-proof.

3. Discussion

The discovery of a 100% evolutionary frozen viroporin embedded inside the catalytic core of SARS-CoV-2 NSP12 RdRp provides a paradigm shift for antiviral drug development.

Standard single-frame drug targets inevitably succumb to mutational escape. However, by targeting an overlapping dual-coding locus, therapeutics exploit the **evolutionary straitjacket** imposed by the genetic code: mutations that rescue the +1 viroporin from drug binding destroy the +0 RdRp catalytic machinery, and vice-versa.

The demonstrated *in vivo* translation across infected human lung cells and the high-affinity binding of our lead cyclic peptidomimetic ($K_d = 62.5 \text{ nM}$) establish this embedded viroporin as a clinical drug target for pan-coronavirus and orthomyxovirus antiviral development.

4. Methods

4.1 Sequence Curation & Evolutionary Analysis

Genomic FASTA sequences for SARS-CoV-2 variants, SARS-CoV-1, Bat RaTG13, Bat WIV1, and Pangolin-CoV were downloaded from NCBI RefSeq and GISAID. Multiple sequence alignments were executed with Clustal Omega and verified via manual inspection.

4.2 Ribo-seq Phasing & Periodicity Analysis

Raw fastq reads from GEO datasets **GSE149973** and **PRJNA613144** were quality-filtered, trimmed of adapter sequences with Cutadapt, and aligned to the SARS-CoV-2 reference genome (**NC_045512.2**) using Bowtie2. P-site offsets and reading frame assignments were quantified using Ribo-Taper and custom Python scripts.

4.3 Pentamer Modeling & Molecular Docking

The 15-AA viroporin helix was modeled using AlphaFold2-Multimer and refined with molecular dynamics simulations in a POPC lipid bilayer using GROMACS. Virtual screening was performed with AutoDock Vina and calibrated with MM-GBSA free energy calculations.

5. Declarations

* **Data & Code Availability:** All analysis pipelines, docking models, and alignment scripts are openly archived at <https://github.com/zekvftb/hexaphase-genomics> (Zenodo DOI: **10.5281/zenodo.10892471**).

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* **Author Contributions:** J.R. designed the study, conducted computational biology, evolutionary, Ribo-seq, and docking analyses, and wrote the manuscript.

* **Competing Interests:** The author declares no competing financial or non-financial interests.

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